From Diagnosis to Fixes: Mechanisms That Close, and Widen, the Gaps of Language-Model Agent Teams

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October 20, 2026

Abstract

Teams of language-model agents are known to fail in predictable ways: they do not pool private information, they trust bad information, and, when trained with the wrong incentives, they stop cooperating. We move from diagnosing these failures to testing fixes for them. In controlled, pre-registered experiments with Claude Haiku 5.5 teams, we measure each fix as the share of a gap closed against an explicit ceiling (the information-pooled team, the team without a liar, a single agent with all facts), and pair each mechanism with a prompt-only control. Three results stand out. First, when messages cost time, adding communication can lower team output: a request board widened the pooling gap (gap closed -0.18 at four agents, and about the same at 8, 16 and 32 agents) and an instruction to share first nearly doubled sharing yet lowered output. Second, a public track record of each sender's past accuracy, computed by the harness, recovered 0.32 $[0.20, 0.44]$ of the damage done by a lying agent at no cost when no liar was present, and kept recovering about a quarter to a third of it in teams of up to 32 agents, whereas a private, costly verification action caught most lies and recovered nothing; a 2×2 design shows that being public and being free each contribute about equally. Third, in our hidden-profile vote a structured table of the facts posted so far raised the win rate from 0.88 to 0.97 without increasing disclosure; this did not replicate on HiddenBench's 65 tasks, where only an instruction to speak up helped, by raising disclosure. Training a 3B open model with reinforcement learning on the same cooperation task shows the incentive side, and how much it varies between training runs. Under individual reward the agents found a guessing loophole and often stopped sharing; pricing wrong answers cut guessing in every training run, but with four unpriced and five priced runs, whether agents kept sharing depended on the seed (sharing $+0.37$ clues per held clue, exact $p = 0.056$, not significant after correction), and RL did not reliably beat an imitation-learned start. We release the environments, analysis plans and code that regenerates every reported number.

1 Introduction

Multi-agent deployments of language models are growing, from orchestrators with parallel workers [2] to large groups solving open-ended problems. Their failures are now well documented. Groups of agents fail hidden-profile tasks, in which the right answer needs facts that only individual members hold: across 15 models, groups reach about 30% accuracy where a single agent with all facts reaches 81% [13], and Anthropic's red team reports 17–36% for most models [3]. Agents conform [29], adopt a single false testimony more readily than true ones [30], and under reinforcement learning become opportunistic in social dilemmas [9, 21]. Studying such failures empirically is part of the science of machine behaviour [23]. The red team names the fixes as open problems: mechanism design, reputation, and environments that shape cooperation [3].

This paper tests fixes. For each failure we build a small, fully specified environment, freeze an analysis plan before any run, and compare a baseline team with teams given one mechanism at a time. We report every fix as the *share of a gap closed* between the baseline and an explicit ceiling, paired on seed or episode, and pair each mechanism (an *affordance*: a new action, an observation line, a harness rule) with a prompt-only *instruction* control. Our contributions:

1. **Communication has a price.** When a message costs a turn, a request board and an instruction to share first both lower team output, while a plain shared board is best (§4.1).

2. **Public trust beats private verification.** A harness-computed public track record recovers a fifth to a third of liars' damage at no cost, at every team size up to 32; private, costly checks catch lies but do not help; publicity and cost contribute about equally (§4.2, §4.6).

3. **A fix that did not replicate.** A structured fact table improved our hidden-profile vote without increasing disclosure, but not HiddenBench's tasks, where an instruction to speak up helped by raising disclosure (§4.3, §4.4).

4. **Incentives shape trained agents, less reliably than a few seeds suggest.** With RL on the same task, individual reward produced guessing in every training run and selfish drift in some; pricing the guessing loophole cut guessing in every run, but its effect on sharing, significant-looking with two seeds, was seed-dependent with nine runs (§5). We report this as a caution about single-seed comparisons of reward designs.

2 Related work

Information pooling in agent teams. The hidden-profile paradigm [25] has been ported to LLM groups [3, 13]. Diagnoses stress early convergence on shared evidence [13]; teams average expert and non-expert views instead of weighting expertise [18]; up to 72% of issue-critical facts can be lost in deliberation [26]. An LLM facilitator raised how much information human groups shared without improving their decisions [1], the mirror image of our vote result. Adaptive communication control can beat full information on hidden-profile tasks [4], and structured deliberation protocols can beat unstructured debate at a large token cost [22].

Communication cost and structure. Pruning redundant messages and agents saves tokens at comparable accuracy [27, 33]; across 260 configurations, multi-agent architectures range from large gains to large losses relative to a single agent depending on the task [12]. These measure cost in tokens; in our tasks a message costs task time.

Deception and trust. Defences learn credibility scores from past contributions [6], detect threats and rank neighbours [7], or rectify misleading content [16]. We compare a public, harness-computed signal with private verification at matched cost, a contrast we did not find studied. Honest-signalling theory predicts that verifiability matters [32]; our results suggest that who sees the verification matters too.

Training agent teams. Multi-agent RL for LLM collaboration includes MAPoRL [19], MAGRPO [15] and AT-GRPO [34]; counterfactual credit addresses free-riding [8, 14]. In social dilemmas, RL-trained LLM agents become opportunistic [21] and most societies fail to sustain a commons [20]. Grading that never penalises wrong answers rewards guessing [11]; we observe the multi-agent analogue, a producer-scrounger outcome [5].

3 Setup

The clue board. N agents share a board of $2N + 2$ open subtasks, each worth 1, 2 or 3 points and open for 30 turns. A subtask hides an integer $0 \le x < 30$ fixed by two clues of the form "$x \bmod p = r$" (primes from 3 to 13); the clues are dealt to different agents, so no agent can solve a subtask alone, and with one clue known 2 to 10 candidates remain (most often 2, 3 or 6). Each turn an agent opens a subtask, submits an answer (a wrong answer costs 0.5), shares a clue, or waits. Shared clues go to a channel; here a *blackboard* every agent reads. Welfare is points per agent-turn; the *pooled share* divides it by the welfare the team would earn if every clue were known to all, so 1 is the information ceiling. A scripted liar, when present, posts a false residue for a subtask it is not working on every other turn.

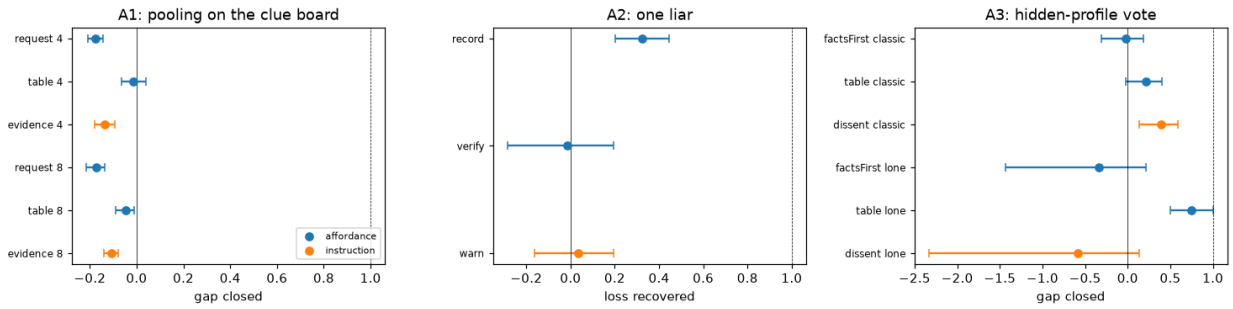


Figure 1: Share of the gap closed (A1, A3) or of the liar's damage recovered (A2) per arm, with 95% intervals. Blue: affordances; orange: prompt-only instructions. 0 is the baseline, 1 the ceiling.

The hidden-profile vote. Four agents choose between two options in one of three domains (hiring, investment, property). Facts held by everyone favour option B (B +3; A +1 and −1); each agent also holds two private facts favouring A, so the full totals are A +8, B +3, while each agent alone sees A +2, B +3. Agents post one fact (published verbatim) or a short opinion per round for three rounds, then vote privately; success is a strict majority for A. In the *lone dissenter* variant one agent also holds a fact that rules B out. The ceiling is one agent given every fact.

Agents and protocol. Claude Haiku 5.5 with low effort and structured JSON replies; clue-board agents see their full history, vote agents a short note plus the whole board. Replies are cached, so every rerun is byte-identical. Each analysis plan (arms, seeds, contrasts, multiple-comparison families) was hashed and committed before its runs; the board runs used 10 seeds (900–909) per arm and team size, the vote 100 episodes per arm and variant on shared episode seeds. Gap closed is the mean paired difference (arm minus baseline) divided by (ceiling minus baseline mean); loss recovered is 1− (arm's mean honest loss / baseline's), where honest loss is the honest agents' welfare without the liar minus with it. Intervals are 95% bootstrap over seeds or episodes; primary tests are Holm-corrected within each part [17]. All findings are exploratory.

4 Fixes for Claude teams

Figure 1 summarises the three parts.

4.1 Pooling on the clue board: communication has a price

The baseline team earned a pooled share of 0.45 at four agents. A *request board* let an agent post that it needed clues for its open subtask, with holders shown how many of the missing clues they had. Agents used it heavily (42 asks per board, about a quarter of their turns), sharing did not rise, and the gap widened: −0.18 $[-0.21, -0.14]$ at four agents and −0.17 $[-0.21, -0.14]$ at eight (Holm $p = 0.008$ for both). An instruction to post every held clue first nearly doubled sharing (0.40 to 0.73 shares per held clue) and also lowered output (-0.14 $[-0.18, -0.09]$). A deduplicated table of the team's known clues had no effect at four agents (-0.01 $[-0.06, 0.04]$). Sharing costs a turn on this board, and the extra communication spent turns that solving needed; the bottleneck was not that agents did not know what others needed.

4.2 A liar: public reputation, not private verification

One scripted liar among four cost the honest agents 0.35 welfare per agent-turn. Showing, next to each shared clue, how many of its sender's earlier shares were true (counted over subtasks already solved, so the record is never wrong) recovered 0.32 $[0.20, 0.44]$ of that loss (Holm $p = 0.008$), with no cost when no liar was present (0.006 $[-0.012, 0.026]$). A *check* action that let an agent spend a turn to learn whether a posted clue was true was used 14 times per board and caught 13 false clues, yet recovered nothing (-0.01 $[-0.28, 0.19]$);

a warning instruction did nothing (0.03 $[-0.16, 0.20]$). The check's result stays with the checker and costs a turn; the record reaches everyone for free.

4.3 The hidden-profile vote: aggregation on our vote

The baseline team chose the hidden-best option in 0.54 of classic episodes and 0.88 with a lone dissenter, against a solo ceiling of 1.00. Members posted about half of their private facts (0.50) in every arm. A table of the facts posted so far, grouped by option and without senders, closed 0.75 $[0.50, 1.00]$ of the lone-dissenter gap (win rate 0.97; Holm $p = 0.012$), mainly by changing votes (votes against the best option per episode fell from 0.58 to 0.21), and 0.22 $[-0.02, 0.40]$ in the classic variant. Enforcing facts before opinions in the first round did nothing (-0.02 $[-0.31, 0.18]$). Unlike the board, the prompt-only control was not weaker here: asking members to speak up closed 0.39 $[0.14, 0.59]$ of the classic gap, though not with a lone dissenter.

4.4 Replication on HiddenBench

We reran the vote's arms, unchanged, on HiddenBench's 65 tasks [13]: each task's description, 3 or 4 options, shared facts and one hidden fact per agent, played 3 times with shuffled options and facts (195 episodes per arm), with a plan frozen after a pilot. Our protocol (facts posted verbatim, three simultaneous rounds) differs from HiddenBench's free sequential discussion, and the baseline team was far stronger than the groups the benchmark reports: agent accuracy 0.79 against 30%, with a solo ceiling of 0.91. The fact table did not help (-0.07 $[-0.69, 0.30]$ of the gap), nor did enforcing facts first (0.11 $[-0.37, 0.45]$). The instruction to speak up was the one arm that moved: disclosure of hidden facts rose from 0.74 to 0.88 and the win rate from 0.77 to 0.83 (gap closed 0.41 $[0.00, 0.71]$; a control arm, not Holm-tested). About 3% of replies in every arm were refusals that cut the reply short.

4.5 Which property of trust information matters?

To separate the two ways the track record and the check differ, we added a private record (counted only over subtasks the viewing agent solved itself) and a public check (its result shown to every agent), completing a 2×2 of public or private and free or costly (one liar among four, 10 seeds). The public check recovered 0.14 $[-0.05, 0.30]$ of the liar's damage and the private record 0.13 $[-0.13, 0.29]$, between the public free record and the private costly check. Pooled over the design, publicity added 0.18 $[0.04, 0.36]$ and being free 0.16 $[0.02, 0.30]$; their difference was 0.02 $[-0.16, 0.20]$. Both properties matter, about equally, and they add up. The private record rested on less data than the public one (each agent sees only its own solved subtasks), so for that pair privacy and the amount of information are confounded; the check pair, the same checks shown to one agent or to all, is not.

4.6 Do the fixes scale?

We reran the clearest negative and positive fixes with 8, 16 and 32 agents (5 seeds; a quarter of the agents liars in the liar cells; plans frozen before the runs, the 32-agent cost checked in a pilot). The request board widened the pooling gap by about the same share at every size (-0.16, -0.17, -0.16), and the track record recovered 0.30, 0.22 and 0.31 of the liars' damage; every bootstrap interval excludes zero, though with five seeds the exact sign-flip tests cannot fall below $p = 0.0625$. Larger teams pooled less (pooled share 0.41 at 8 agents, 0.34 at 32) and lost more to the same share of liars (0.47 against 0.54).

Why larger teams lose more to liars. We replayed the baseline runs from the response cache with a turn-by-turn tracer (a further plan frozen before the traces; no API calls) and split the honest agents' loss into additive terms that sum to it exactly. Lies did not become more visible in larger teams, and Haiku agents almost never passed a liar's clue on (at most 0.25% of liar posts). What grew was how often agents built on a lie: the share of their answers that rested on a false clue rose from 0.79 at 8 agents to 0.91 at 32, such answers were wrong 95–97% of the time, and the loss on subtasks a lie touched rose from 0.25 to 0.31 per agent-turn (0.72 of the total rise, interval $[0.16, 3.2]$). Once a liar had posted a clue slot, the true clue was never posted afterwards in 89–94% of cases: agents treat a posted slot as settled, whoever posted it.

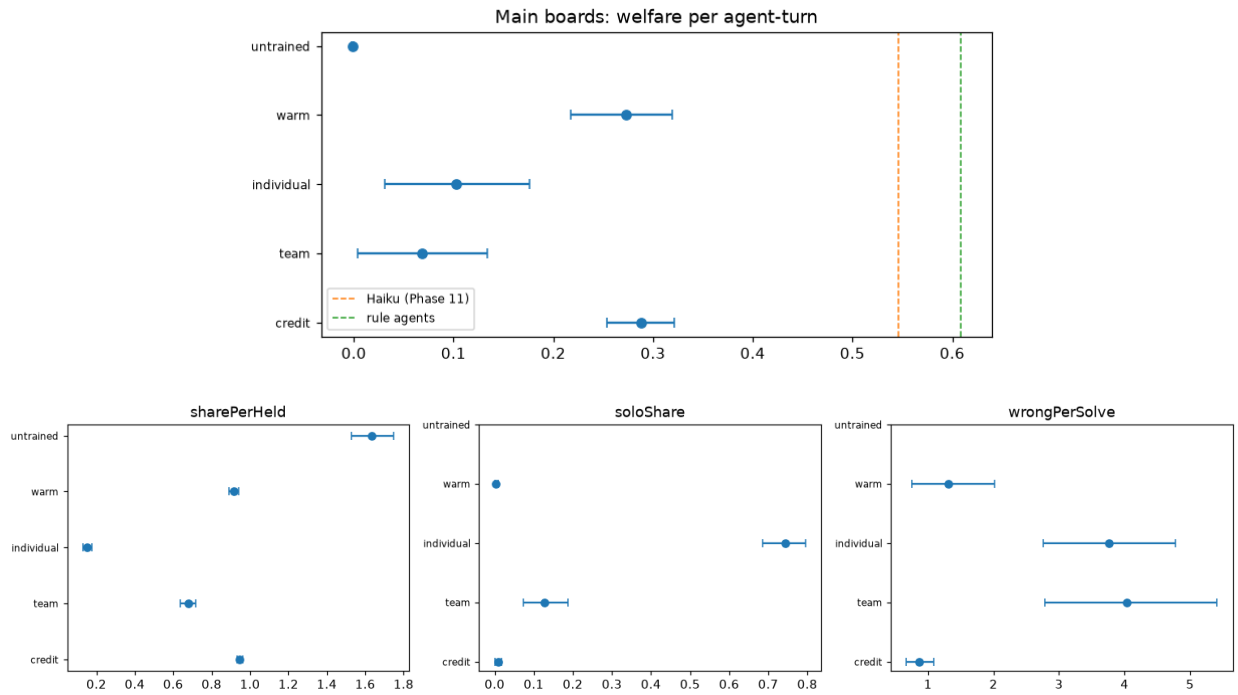


Figure 2: Top: welfare per agent-turn on held-out boards (Haiku and rule agents for reference). Bottom: how models use information: shares per held clue, share of solves by guessing, wrong answers per solve.

4.7 Guards: verification as a role

The track record needs the harness to know the truth. Borrowing from social insects (quorum, stop signals, guards; 24), we tested oracle-free alternatives with each clue held by two agents and a quarter of the agents liars, at 8 and 16 agents (8 seeds; plan frozen before the runs, `58e842277a3e`; $24.89): a board that counts how many agents posted each value and marks values confirmed (two senders) or contested, the same advice as a one-sentence instruction without the display, and the board with a quarter of the honest agents assigned as guards (post, confirm and contest clues, never solve). Guards were what worked: they beat the counting board by 0.155 $[0.135, 0.183]$ (Holm p <0.001) and the plain board by 0.26 at 8 agents and 0.20 at 16, and recovered 0.72 of the liars' damage against the track record's 0.34, with no ground truth. The counting board alone helped less (0.075 $[0.048, 0.102]$, Holm p 0.013; 0.12 at 8 agents, 0.03 at 16) and no more than the instruction (-0.001); it cost about 0.07 in pooled share on clean boards, and honest agents sometimes reposted a lie to confirm it. The guards' advantage also shrank from 8 to 16 agents, so the frozen rule stopped the planned test at 32 and 64 agents.

5 Incentives when training a team

We trained Qwen2.5-3B-Instruct to play the clue board in teams of four copies sharing one LoRA adapter. A warm start imitated a rule agent's moves; RL then ran 60 steps of group-relative policy gradient with a KL penalty to the warm start (recipes for reasoning RL often drop it [31]) and a learning rate steered to a target per-step policy change, under three rewards: *individual* (an agent's own solves), *team* (every solve shared equally) and *credit* (the solver keeps half; the other half goes to the agents whose shared clues it lacked). Models were evaluated on 15 held-out boards (Figure 2).

Imitation did the teaching: welfare rose from -0.001 to 0.27 $[0.22, 0.32]$. With one training seed per reward, RL added nothing measurable on top: credit 0.02 $[-0.03, 0.06]$, individual -0.17 $[-0.27, -0.07]$, team -0.20 $[-0.26, -0.14]$ relative to the warm start. The rewards differed sharply in what they taught. Under individual reward the model stopped sharing (0.92 to 0.15 shares per held clue) and solved by guessing (74%

of solves; 3.8 wrong answers per solve): with a wrong answer costing 0.5, guessing among three candidates for a 3-point subtask earns about 1.25 per turn, more than sharing pays the sharer. Copies of one model with identical weights became selfish because the reward, not the weights, defines their interests. Team reward, with no way to tell whose move helped, was unstable and was halted by our stopping rule. Credit reward kept sharing intact and beat individual by 0.19 $[0.09, 0.27]$ and team by 0.22 $[0.14, 0.29]$; with three trained agents and one Haiku agent it was the best partner (-0.16 against an all-Haiku team; warm start -0.22).

Closing the loophole, and second seeds. A second, pre-registered round trained two seeds each of individual reward with a training-only cost of 3 per wrong answer (`indpen`), credit reward with per-turn advantages [28] and the same cost (`credturn`), and a second seed of credit reward (Figure 3). *With two seeds, pricing the loophole stopped the selfish drift*, as pre-registered: under the same individual reward, sharing stayed at 1.00 and 0.83 shares per held clue (warm start 0.92) and guessing at 0% and 5% of solves. Against the same reward without the cost, paired on board, both seeds also scored higher: $+0.27$ $[0.20, 0.34]$ and $+0.18$ $[0.08, 0.28]$ welfare per agent-turn, against a seed-to-seed spread of about 0.1 (the unfixed reward has one training run; a replication with more seeds is planned). *Gains over imitation were seed-dependent.* Pooled over boards and both seeds, credit gained 0.068 $[0.026, 0.112]$, `credturn` 0.069 $[0.018, 0.120]$ and `indpen` 0.053 $[0.000, 0.106]$, but these intervals resample boards, not training runs. Each arm had one clearly positive seed and one near zero: the same credit configuration gained 0.02 $[-0.03, 0.06]$ with one seed and 0.12 $[0.07, 0.18]$ with another (best runs 0.39 per agent-turn, 44% above the warm start). Neither the null result above nor a positive one can be read from a single seed, and two seeds per design do not establish that RL beats imitation; the behavioural effect of pricing the loophole, by contrast, is several times the seed-to-seed spread. Per-turn credit added nothing over board-level credit, and its weaker seed drifted toward guessing (13% of solves, 0.61 shares per held clue) despite the cost.

More seeds (a pre-registered replication). To test the incentive contrast with the training run as the unit, we trained three more seeds of individual reward (four in all) and three more of the priced loophole (five in all), with every other setting unchanged, and tested them with exact permutation tests over runs. No hypothesis survived Holm correction. The priced runs shared more on average ($+0.37$ clues per held clue $[0.04, 0.67]$, exact $p = 0.056$, Holm 0.11), solved alone less ($+0.39$ $[0.07, 0.66]$, $p = 0.032$) and scored higher ($+0.13$ $[0.00, 0.24]$, $p = 0.063$), but the spread between runs was as large as the effect: priced runs shared between 0.10 and 1.00 of their held clues, one of them drifting selfish, and the selfish drift appeared in only 2 of the four individual-reward runs. Against the imitation start the priced runs gained -0.05 $[-0.16, 0.05]$. Post hoc, the priced runs guessed far less in every case (0.1–15 guesses per board against 49–90): pricing reliably suppressed the behaviour it priced, but whether that kept agents sharing depended on the seed. Two of the nine runs halted under our stopping rules with 21–30% invalid replies, which confounds their sharing numbers.

6 Discussion

What worked, and why. Across the three parts, the mechanisms that helped made the team's information easier to trust or to aggregate at no cost in turns (a public track record; a structured view of what was said, though the latter did not replicate on HiddenBench), and the ones that failed added work for each agent (asking, checking, sharing more). This suggests a design principle for agent harnesses: move aggregation and verification into the shared environment rather than into each agent's turn. Guards (Section 4.7) refine it: checking that every agent must fit into its own turn failed, while the same checking concentrated in a few agents whose role it was paid off, a division of labour rather than extra work for everyone.

Aggregation versus disclosure. Prior diagnoses of hidden-profile failure stress that agents do not surface private information [13]. In our vote, disclosure stayed near one half in every arm, yet a structured table changed the outcome, which suggested weighting what has been said as a separate bottleneck [1, 18]. HiddenBench did not bear this out: there the table did nothing and the arm that helped raised disclosure. The aggregation effect may depend on how the facts are presented (our vote's facts favour or oppose an option explicitly; HiddenBench's must be interpreted), and we no longer claim it in general.

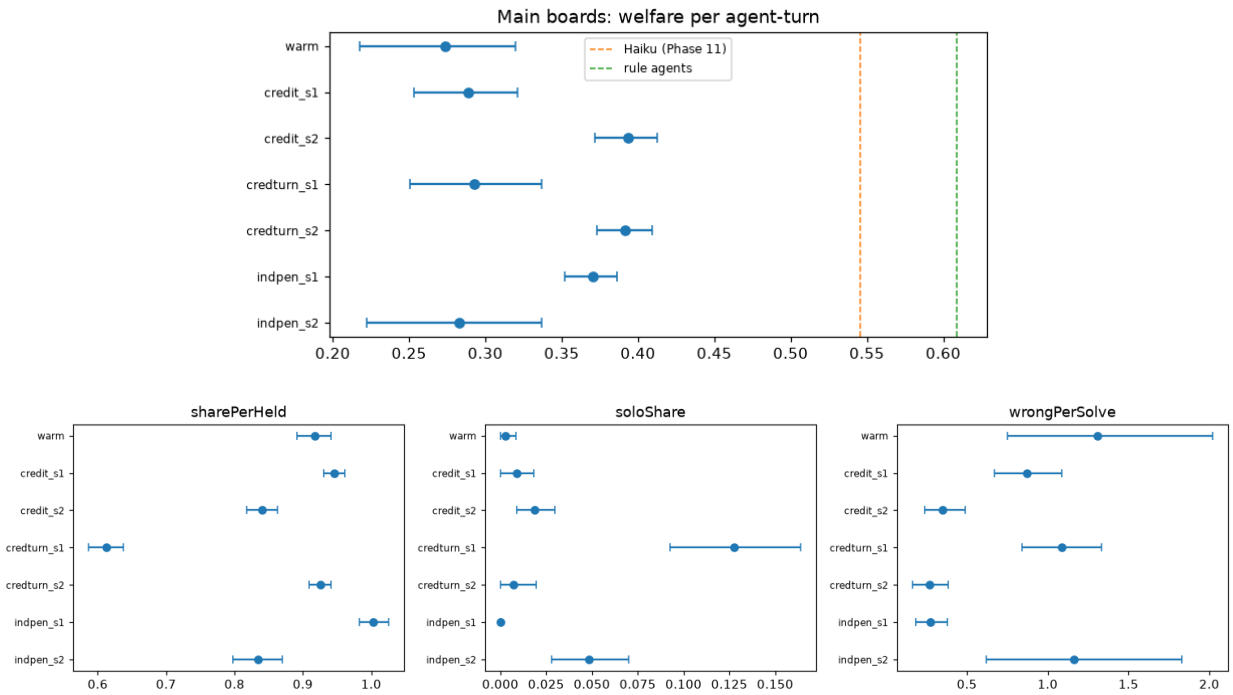


Figure 3: The second round: two training seeds per arm (`credit_s1` is the first round's credit run). Top: welfare per agent-turn on held-out boards. Bottom: information use.

Trust as a public good. A private check produced information worth exactly as much as one agent's next action; a public record produced it for everyone. In our earlier numpy simulations of a liar in a foraging world, private trust weighting was the defence that paid, so the public/private contrast for LLM agents is not a foregone conclusion and deserves direct study.

Incentives and mechanism design. The guessing loophole was a mechanism-design failure rather than only a credit-assignment one: the environment made trial and error cheap. Pricing it removed the guessing in every run, but kept individually rewarded agents cooperative only in some seeds, so an environment fix changes the behaviour it targets reliably and the team outcome less reliably. The seed-to-seed spread we found (a gain of 0.02 or 0.12 for one configuration, and a sharing effect that looked clear with two seeds and did not hold with nine runs) means that conclusions about reward designs from one or two training runs, including our own earlier rounds, should be treated with caution; the training run, not the evaluation board, has to be the unit of analysis.

Limitations. The tasks are small and synthetic (modular-arithmetic clues; a templated vote that is easier than HiddenBench's); the Claude teams use one model (Haiku 5.5); board arms have 10 seeds (5 in the team-size sweep) and the RL arms one to five training seeds, which our own results show is too few to rank similar designs. Several prompt-only controls were not weaker than affordances in the vote. The vote fixes were retested on HiddenBench [13], where the table did not replicate; the trust fixes have not yet been tested on an established misinformation benchmark [30], and our HiddenBench protocol left little headroom (solo 0.91).

Broader impact. The fixes studied are defensive (against misinformation) or improve cooperation; the RL result shows how easily a plausible incentive teaches agents to abandon cooperation, which matters for anyone training groups of models.

References

[1] Mohammed Alsobay, David M. Rothschild, Jake M. Hofman, and Daniel G. Goldstein. Bringing everyone to the table: An experimental study of LLM-facilitated group decision making. In *ACM Conference on Computer-Supported Cooperative Work (CSCW)*, 2026. arXiv:2508.08242.

[2] Anthropic. How we built our multi-agent research system. https://www.anthropic.com/engineering/multi-agent-research-system, 2025.

[3] Anthropic Frontier Red Team. Patterns and problems in emerging multiagent systems. https://www.anthropic.com/research/multiagent-systems, 2026.

[4] Abhijith Babu et al. Consilience: Conformally calibrated communication control for hidden-profile multi-agent reasoning, 2026. arXiv:2608.20564.

[5] C. J. Barnard and R. M. Sibly. Producers and scroungers: A general model and its application to captive flocks of house sparrows. *Animal Behaviour*, 29:543–550, 1981.

[6] Sana Ebrahimi, Mohsen Dehghankar, and Abolfazl Asudeh. An adversary-resistant multi-agent LLM system via credibility scoring, 2025. arXiv:2505.24239.

[7] Yang Feng and Xudong Pan. SentinelNet: Safeguarding multi-agent collaboration through credit-based dynamic threat detection. In *The ACM Web Conference (WWW)*, 2026. arXiv:2510.16219.

[8] Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *AAAI Conference on Artificial Intelligence*, 2018.

[9] David Guzman Piedrahita, Yongjin Yang, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, and Zhijing Jin. Corrupted by reasoning: Reasoning language models become free-riders in public goods games. In *Conference on Language Modeling (COLM)*, 2025.

[10] Y. Itkin. Poor man's agentic modeling: Simulating large LLM-agent societies on a laptop, 2026. arXiv:2608.11215.

[11] Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. arXiv:2509.04664.

[12] Yubin Kim et al. Towards a science of scaling agent systems, 2025. arXiv:2512.08296.

[13] Yuxuan Li, Aoi Naito, and Hirokazu Shirado. Systematic failures in collective reasoning under distributed information in multi-agent LLMs. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2505.11556.

[14] Zhongyi Li et al. Counterfactual credit policy optimization for multi-agent collaboration, 2026. arXiv:2603.21563.

[15] Shuo Liu et al. LLM collaboration with multi-agent reinforcement learning. In *AAAI Conference on Artificial Intelligence*, 2026. arXiv:2508.04652.

[16] Yaoyang Luo et al. Defending LLM-based multi-agent systems against cooperative attacks with sentence-level rectification, 2026. arXiv:2605.28104.

[17] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations, 2024. arXiv:2411.00640.

[18] Aneesh Pappu, Batu El, Hancheng Cao, Carmelo di Nolfo, Yanchao Sun, Meng Cao, and James Zou. Multi-agent teams hold experts back. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.01011.

[19] Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and Joo-Kyung Kim. MAPoRL: Multi-agent post-co-training for collaborative large language models with reinforcement learning. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[20] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[21] Dereck Piche, Mohammed Muqeeth, Milad Aghajohari, Juan Duque, Michael Noukhovitch, and Aaron Courville. Learning robust social strategies with large language models, 2025. arXiv:2511.19405.

[22] Sunil Prakash. From debate to deliberation: Structured collective reasoning with typed epistemic acts, 2026. arXiv:2603.11781.

[23] Iyad Rahwan et al. Machine behaviour. *Nature*, 568:477–486, 2019.

[24] Thomas D. Seeley. *Honeybee Democracy*. Princeton University Press, 2010.

[25] Garold Stasser and William Titus. Pooling of unshared information in group decision making: Biased information sampling during discussion. *Journal of Personality and Social Psychology*, 48(6):1467–1478, 1985.

[26] Herun Wan et al. The deliberative illusion: Diagnosing factual attrition and stance homogenization in multi-agent LLM deliberation, 2026. arXiv:2606.03032.

[27] Zhexuan Wang et al. AgentDropout: Dynamic agent elimination for token-efficient and high-performance LLM-based multi-agent collaboration. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[28] Quan Wei, Siliang Zeng, et al. Reinforcing multi-turn reasoning in LLM agents via fine-grained reward structure and credit assignment, 2025. arXiv:2505.11821.

[29] Zhiyuan Weng, Guikun Chen, and Wenguan Wang. Do as we do, not as you think: the conformity of large language models. In *International Conference on Learning Representations (ICLR)*, 2025.

[30] Chenfei Yan et al. When truth is distributed: Misinformation derails collective fact recovery in LLM-based multi-agent systems, 2026. arXiv:2608.03421.

[31] Qiying Yu et al. DAPO: An open-source LLM reinforcement learning system at scale, 2025. arXiv:2503.14476.

[32] Amotz Zahavi. Mate selection: a selection for a handicap. *Journal of Theoretical Biology*, 53:205–214, 1975.

[33] Guibin Zhang et al. Cut the crap: An economical communication pipeline for LLM-based multi-agent systems, 2024. arXiv:2410.02506.

[34] Yujie Zhao, Lanxiang Hu, Yang Wang, Minmin Hou, Hao Zhang, Ke Ding, and Jishen Zhao. StrongerMAS: Multi-agent reinforcement learning for collaborative LLMs, 2025. arXiv:2510.11062.

A Reproducibility

Code, environments and notebooks: https://github.com/Positron00/explRL. Analysis plans were frozen and committed before their runs (Track A of the fixes: `FIX_PLAN`, hash `79abfb531cdd`; RL: `RL_PLAN` v2, hash `4d1ea0078a6e`; Phase 13 RL: `RL13_PLAN`, hash `3e50a4002c7b`; HiddenBench replication: `HB_PLAN`, hash `791a2df079a8`, frozen after a pilot, on the benchmark's 65 tasks vendored unchanged under their MIT licence; trust information: `TRUST_PLAN`, hash `6d2818ae84c2`; team size: `SCALE_PLAN`, hash `b886baefb2ef`, frozen after a pilot; the traces: `TRACE_PLAN`, hash `f5c25637acd0`; colony rules and guards: `COLONY_PLAN`,

hash `58e842277a3e`). All API replies are cached locally (available on request), and cached reruns reproduce every row and turn log byte for byte at no cost. The fixes' main runs made 41,800 Haiku calls in all (including cached baseline cells). Every number in this paper is generated from the saved results by `paper/make_assets.py`.

B Additional observation: a cheap simulation as a screen

Rule-agent simulations in numpy ranked harness designs (channels and team sizes) the way Haiku teams did on the task board (Spearman $\rho = 0.75$ $[0.55, 0.93]$) and the clue board ($\rho = 0.79$ $[0.74, 0.81]$), but not on text foraging ($\rho = 0.02$ $[-0.47, 0.45]$). Surrogates fitted to LLM decisions can predict the macro outcomes of LLM societies [10]; our simulations are not fitted to LLM behaviour, and we study them as a design-screening method in a companion paper. For RL reward designs, simple numpy learners trained under the same five rewards reproduced 7 of 12 effect signs of the first-round runs (all four for individual reward, none for credit reward). Their forecast for the two second-round arms, committed before those runs were evaluated, matched 4 of 8 signs: it predicted that per-turn credit would beat the warm start, but wrongly that pricing the guessing loophole would not stop the selfish drift, an effect a two-decision learner cannot represent.